

```
int senpin=A7;
int senval=0;

void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
  Serial1.begin(9600);
}


```

Fig. 5: Transmitter code1

```
void loop() {
  senval=analogRead(senpin);
  Serial.println(senval);
  delay(100);
  // put your main code here, to run repeatedly:
}


```

Fig. 6: Transmitter code for updating value from EMG sensor

```
#include<Servo.h>
Servo arm;

int recvvval=0;

void setup() {
  // put your setup code here, to run once:
  arm.attach(9);
  Serial1.begin(9600);
}


```

Fig. 7: Connection for receiver

```
void loop() {
  recvvval= Serial.parseInt();
  // put your main code here, to run repeatedly:
  int pos= map(recvvval,60,400, 0, 180);
  arm.write(pos);
}


```

Fig. 8: Receiver code loop function

prosthetic hand data transmitter and a prosthetic hand data receiver with servo move mechanism. For each of them we need to create separate codes.

The code for transmitter is required for EMG data transmission to the prosthetic hand. For that we first define the pin numbers for the sensor and variables to store the EMG sensor data. Then we create a setup function where the baud rate is set for the Bluetooth and serial port

communication.

The Arduino Pro Micro offers two hardware serial ports—one for USB and the other for receiver and transmitter pins. Use Serial1 for Bluetooth communication by setting its baud rate to 9600, which is the default for Bluetooth HC-05. Then create a loop function (see Fig. 6) for sending updated EMG sensor values to the Bluetooth in a regular interval of time.

For receiver, include the servo

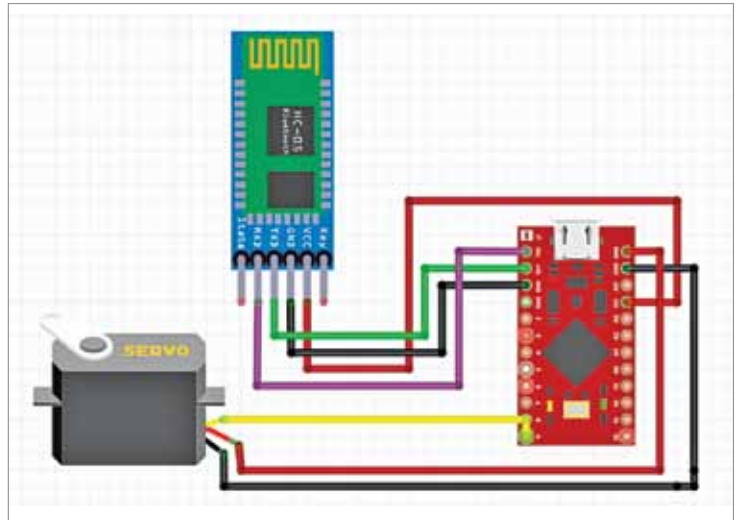


Fig. 9: Receiver circuit connections

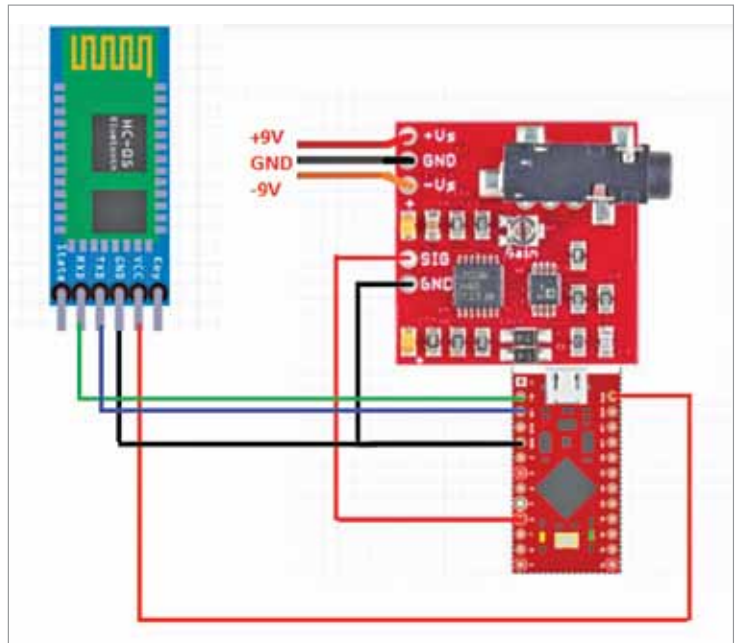


Fig. 10: Transmitter circuit connections

library servo.h in the code to control it. Then create a setup function to set the pin number for connecting the PWM signal wire of servo to control it.

Arduino Pro Micro has several PWM pins; we will use pin number 9 for servo PWM control. Start the serial to read the transmitter device value and then create a loop function to get the EMG sensor value over serial. Apply the EMG value to the servo degree rotation range,